

Graphs of parabolas

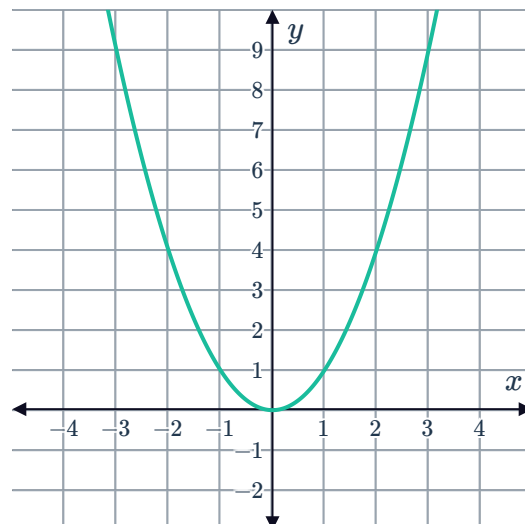


1

a

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

b



c No

d $x = 0$

e 0

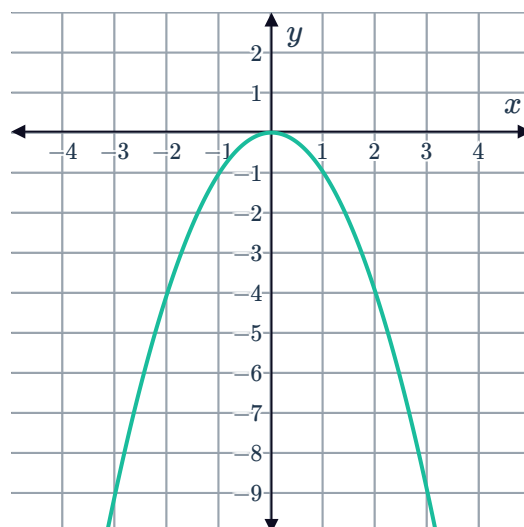
f 2

2

a

x	-3	-2	-1	0	1	2	3
y	-9	-4	-1	0	-1	-4	-9

b



c No

d $x = 0$

e 0

3

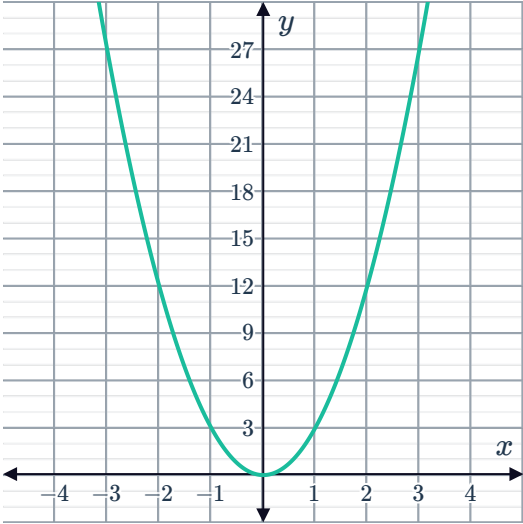
a

i



x	-3	-2	-1	0	1	2	3
y	27	12	3	0	3	12	27

ii

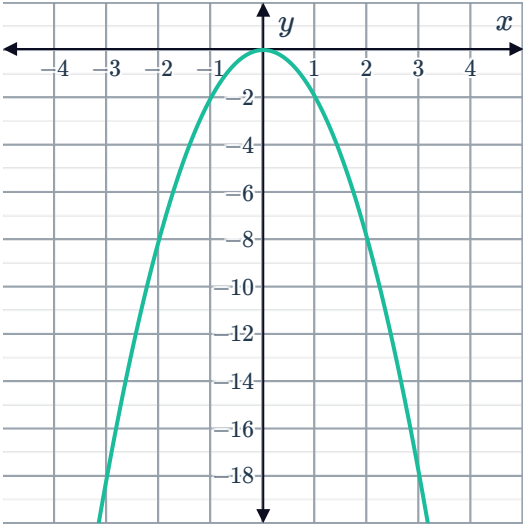


b

i

x	-3	-2	-1	0	1	2	3
y	-18	-8	-2	0	-2	-8	-18

ii

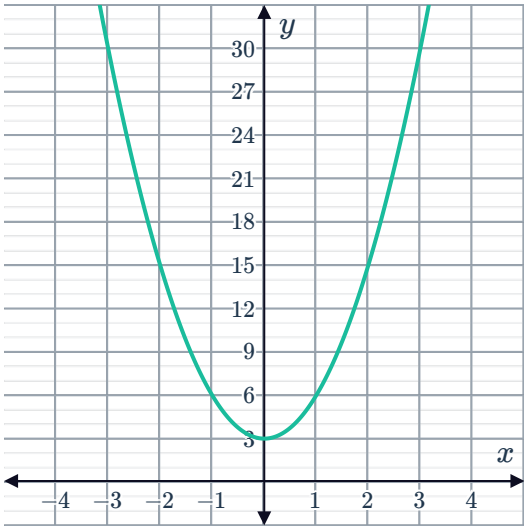


c
i



x	-3	-2	-1	0	1	2	3
y	30	15	6	3	6	15	30

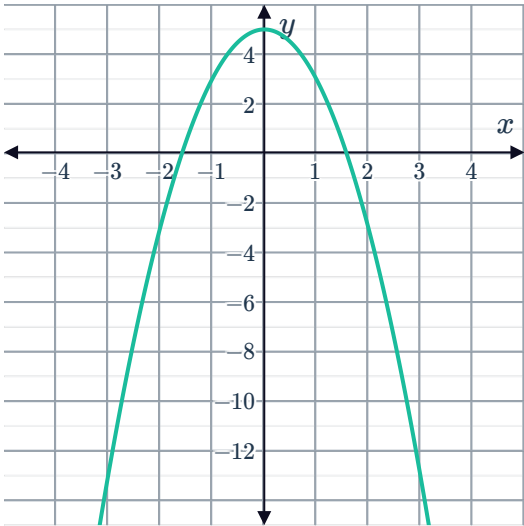
ii



d
i

x	-3	-2	-1	0	1	2	3
y	-13	-3	3	5	3	-3	-13

ii



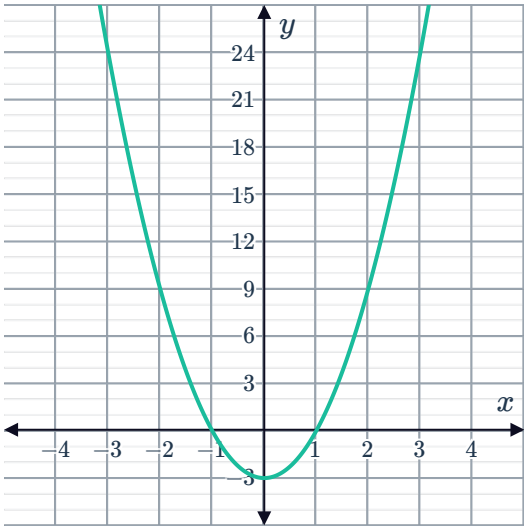
e

i



x	-3	-2	-1	0	1	2	3
y	24	9	0	-3	0	9	24

ii

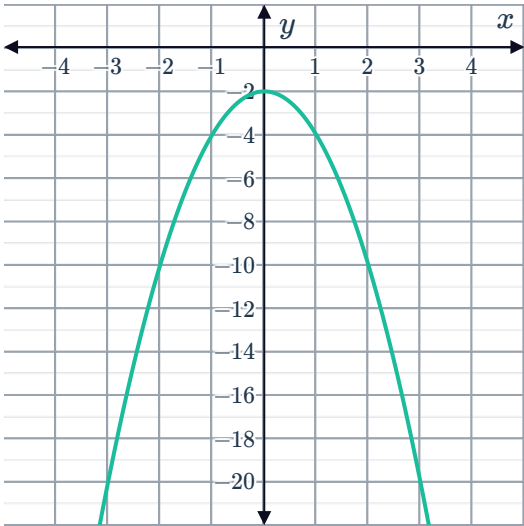


f

i

x	-3	-2	-1	0	1	2	3
y	-20	-10	-4	-2	-4	-10	-20

ii



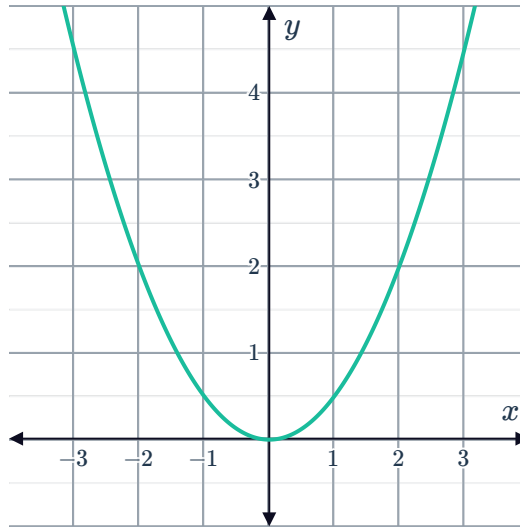
g

i



x	-3	-2	-1	0	1	2	3
y	$\frac{9}{2}$	2	$\frac{1}{2}$	0	$\frac{1}{2}$	2	$\frac{9}{2}$

ii

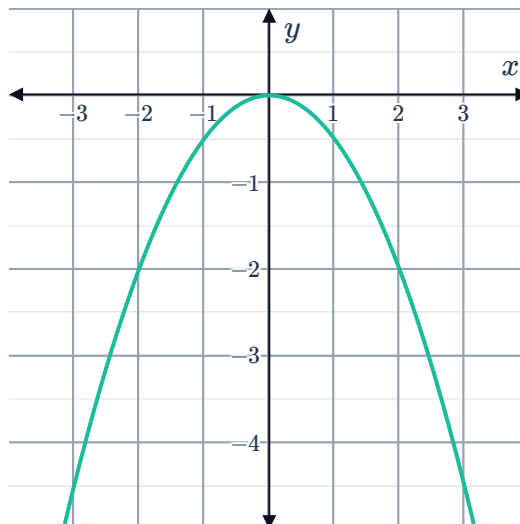


h

i

x	-3	-2	-1	0	1	2	3
y	$-\frac{9}{2}$	-2	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-2	$-\frac{9}{2}$

ii





Transformations of parabolas

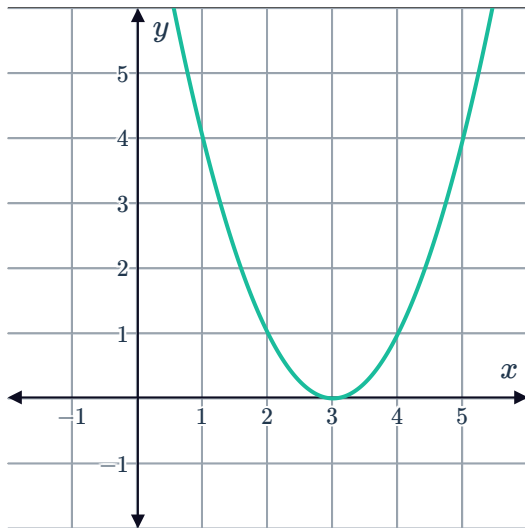
4

a

i

x	1	2	3	4	5
y	4	1	0	1	4

ii



iii 0

iv $x = 3$

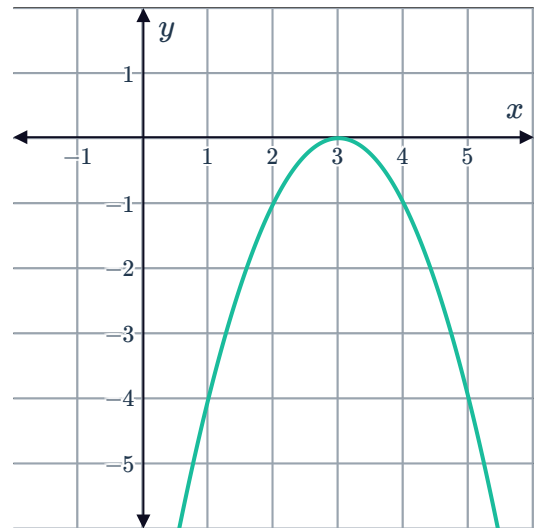
v (3, 0)

b

i

x	1	2	3	4	5
y	-4	-1	0	-1	-4

ii



iii 0

iv 3

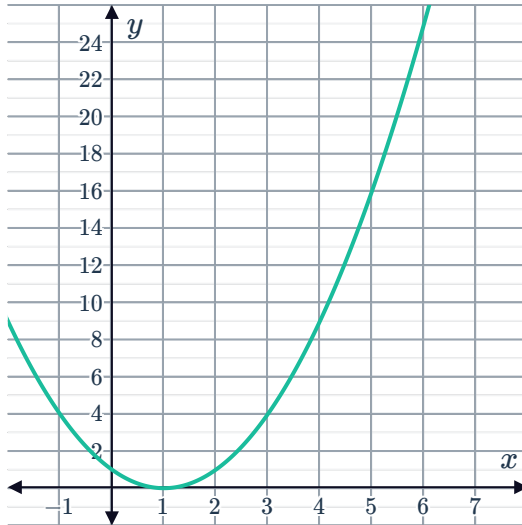
v (3, 0)

c

i

x	1	2	3	4	5
y	0	1	4	9	16

ii



iii 0

iv 1

v (1, 0)

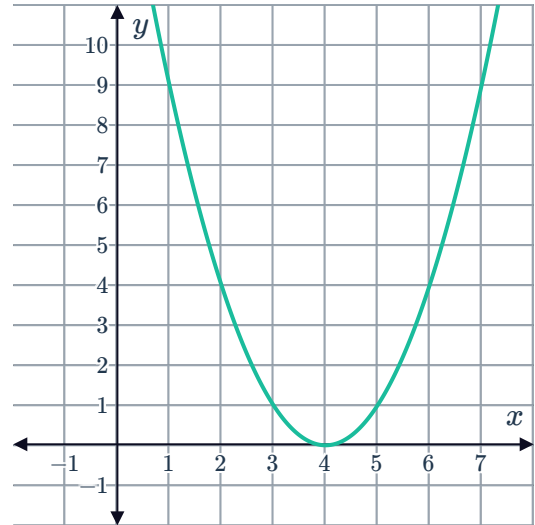


d

i

x	1	2	3	4	5
y	9	4	1	0	1

ii



iii 0

iv 4

v (4, 0)

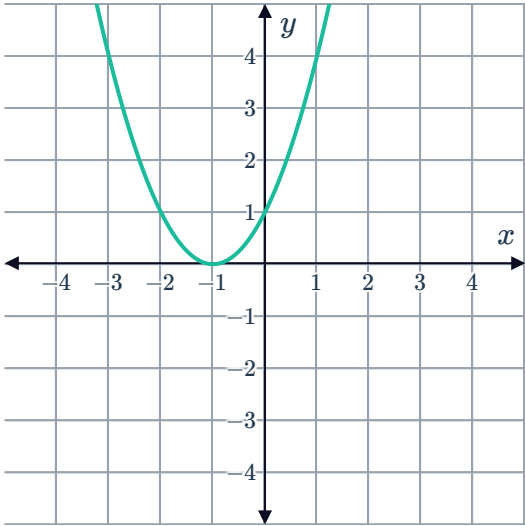
5



a
i

x	-3	-2	-1	0	1
y	4	1	0	1	4

ii

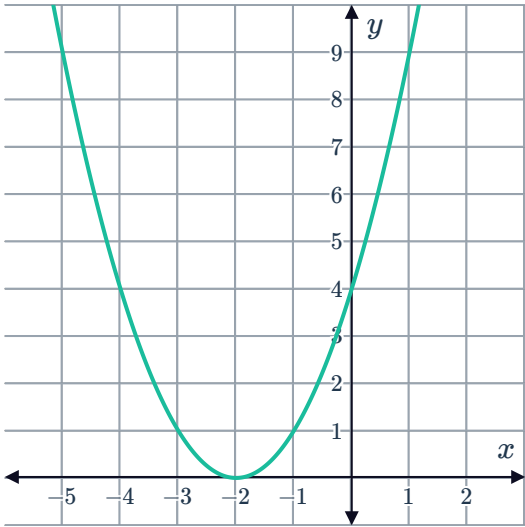


- iii 0
iv $x = -1$
v $(-1, 0)$

c
i

x	-3	-2	-1	0	1
y	1	0	1	4	9

ii

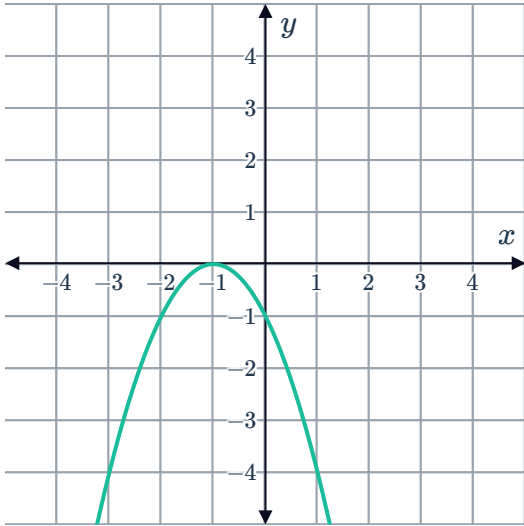


- iii 0
iv -2
v $(-2, 0)$

b
i

x	-3	-2	-1	0	1
y	-4	-1	0	-1	-4

ii

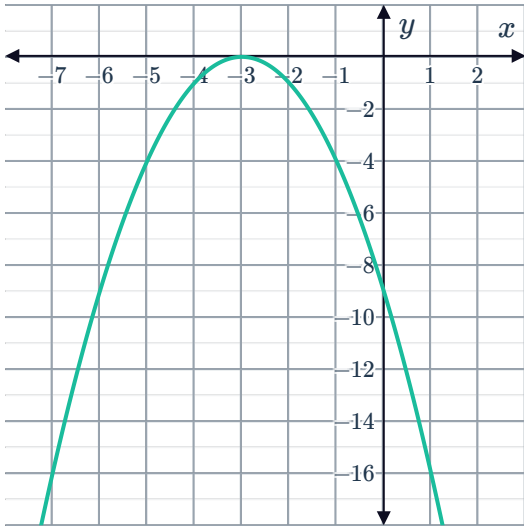


- iii 0
iv -1
v $(-1, 0)$

d
i

x	-3	-2	-1	0	1
y	0	-1	-4	-9	-16

ii



- iii 0
iv -3
v $(-3, 0)$

6

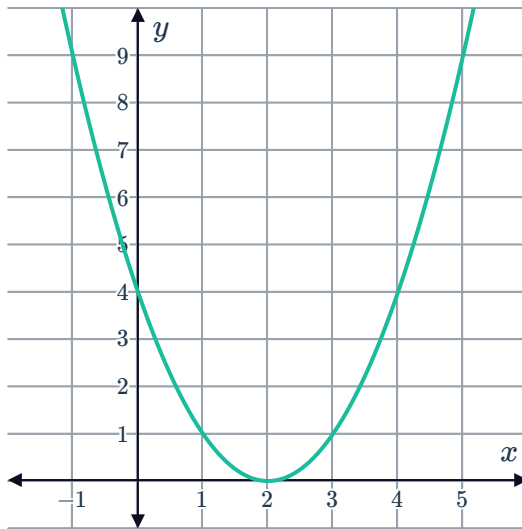


a

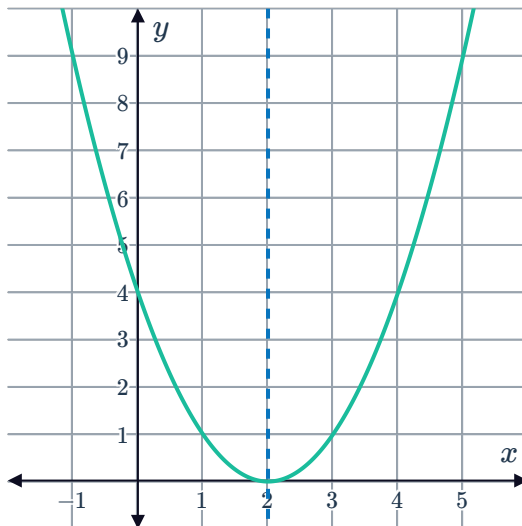
i $(2, 0)$

ii $x = 2$

iii



iv

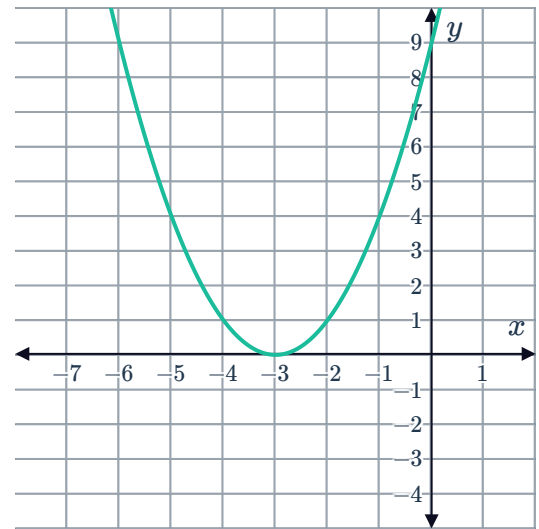


b

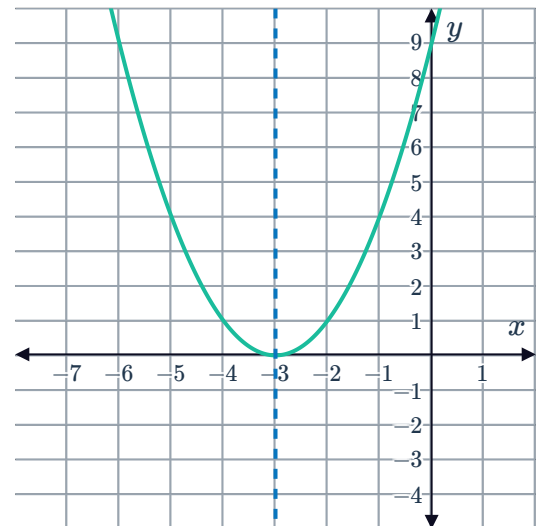
i $(-3, 0)$

ii $x = -3$

iii



iv

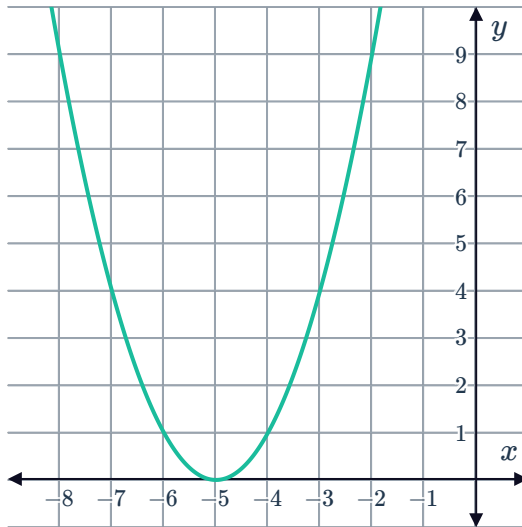


c

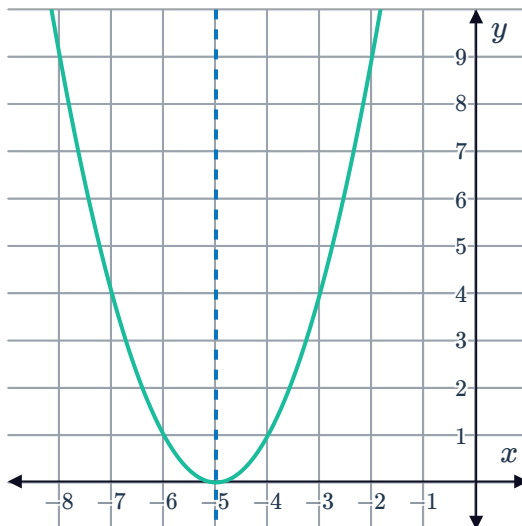
i $(-5, 0)$

ii $x = -5$

iii



iv

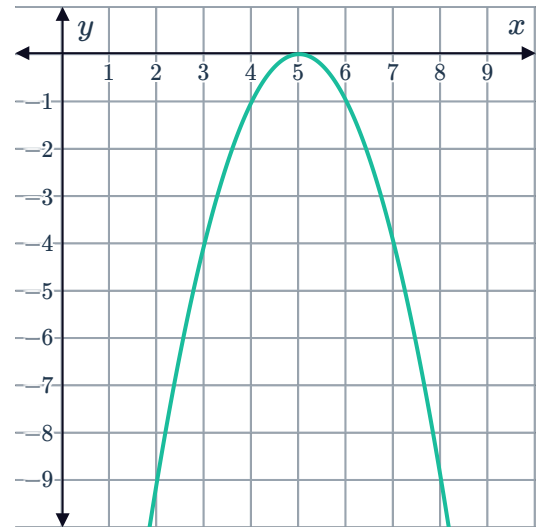


d

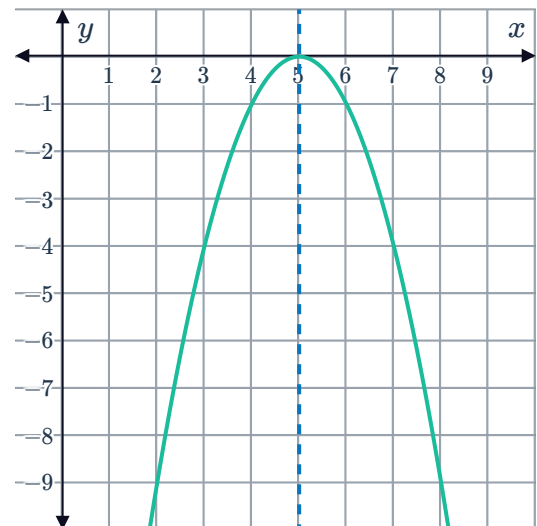
i $(5, 0)$

ii $x = 5$

iii



iv

7 Horizontally translated by -4 units.

8

a Concave up

b Concave down

c Concave down

d Concave up

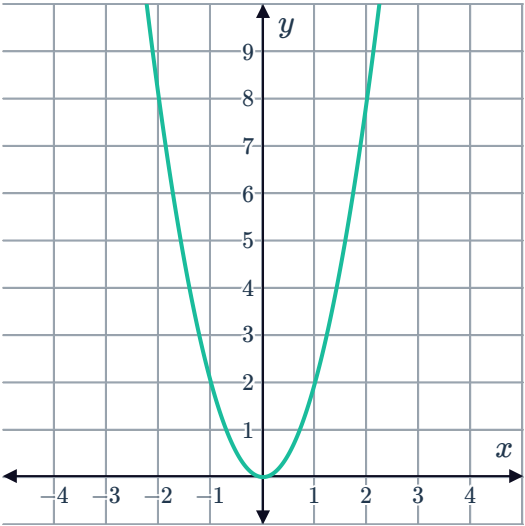
9

a

i

x	-2	-1	0	1	2
y	8	2	0	2	8

ii



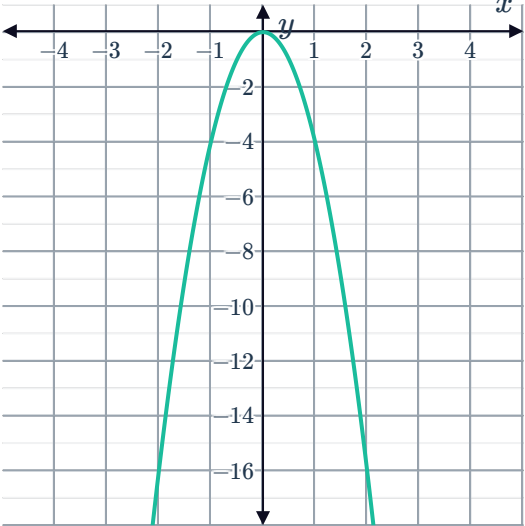
iii (0,0)

c

i

x	-2	-1	0	1	2
y	-16	-4	0	-4	-16

ii



iii (0,0)

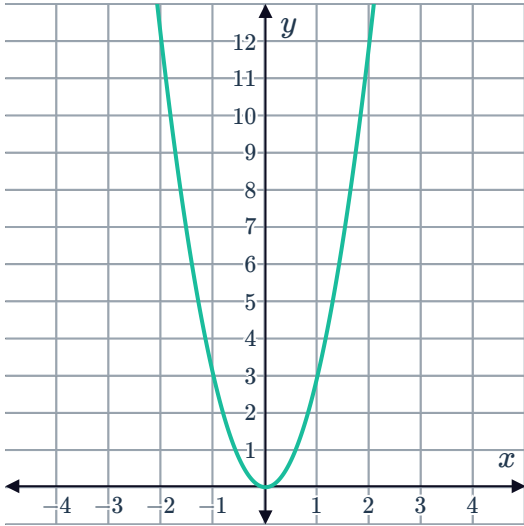


b

i

x	-2	-1	0	1	2
y	12	3	0	3	12

ii



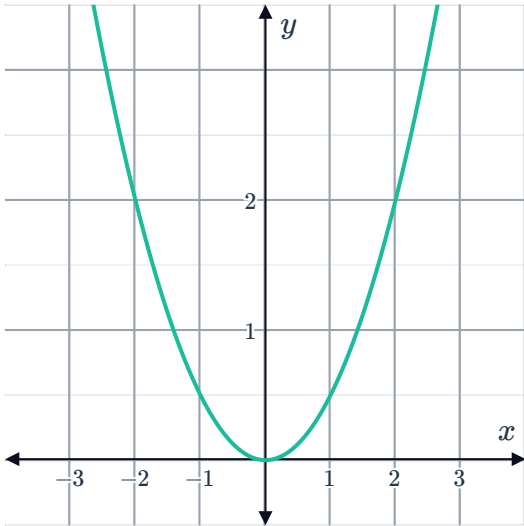
iii (0,0)

d

i

x	-2	-1	0	1	2
y	2	$\frac{1}{2}$	0	$\frac{1}{2}$	2

ii



iii (0,0)

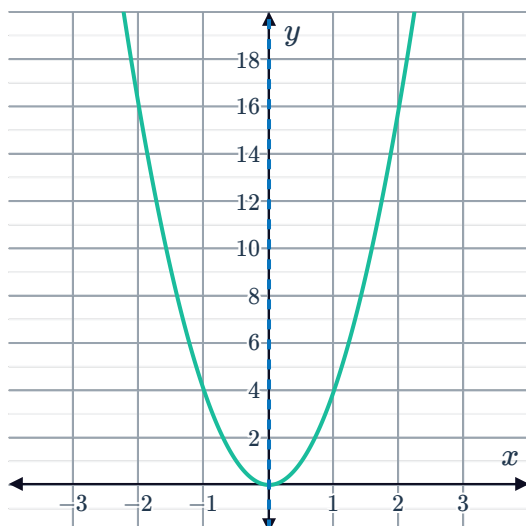
10



a

i $x = 0$

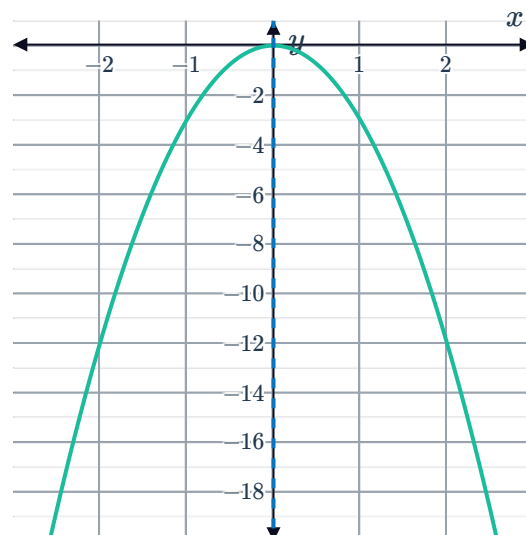
ii



b

i $x = 0$

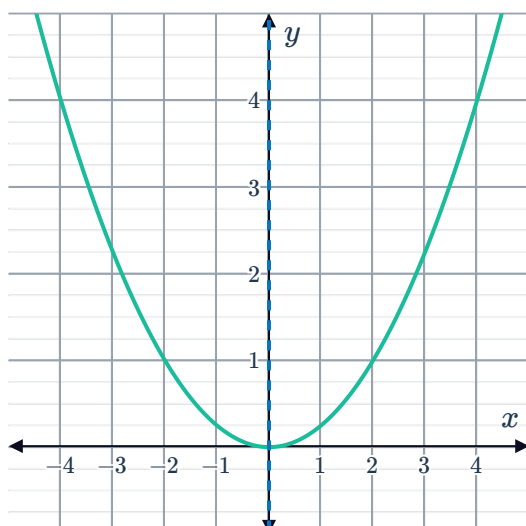
ii



c

i $x = 0$

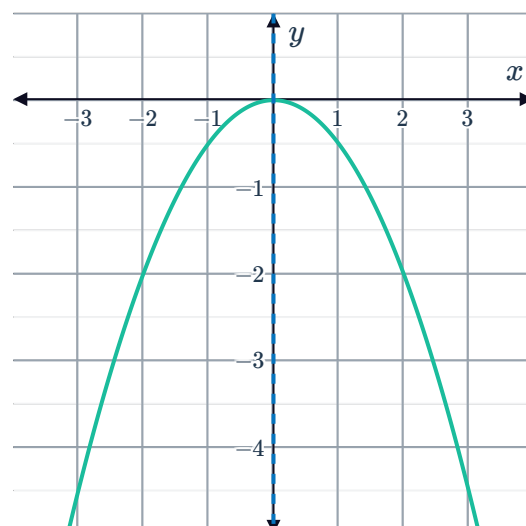
ii



d

i $x = 0$

ii



11 Vertically scaled by 5 units.

12

a $y = x^2 + 3$

b $y = x^2 - 2$

c $y = (x - 5)^2$

d $y = (x + 4)^2$

e $y = 2x^2$

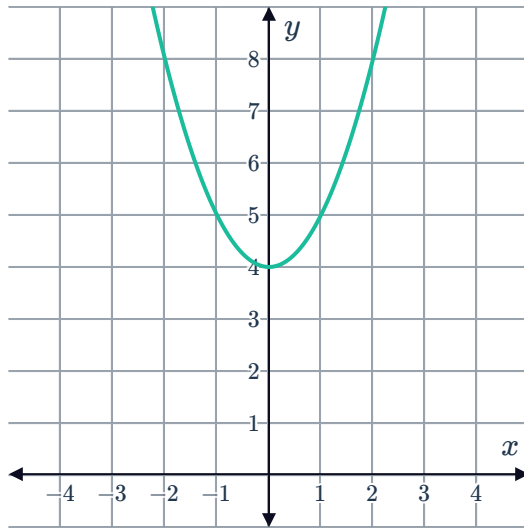
f $y = -3x^2$

g $y = \frac{1}{2}x^2$

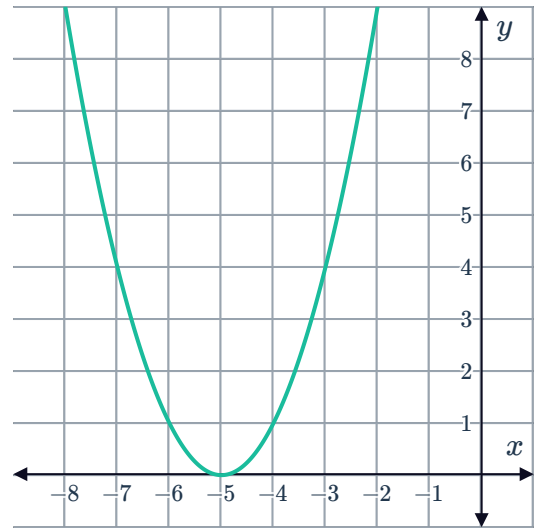
h $y = -x^2$

13

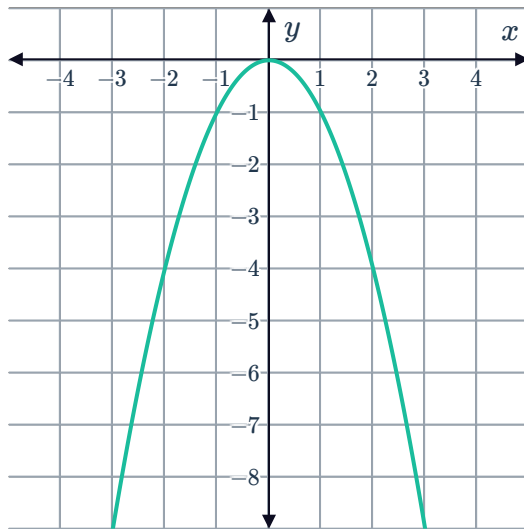
a



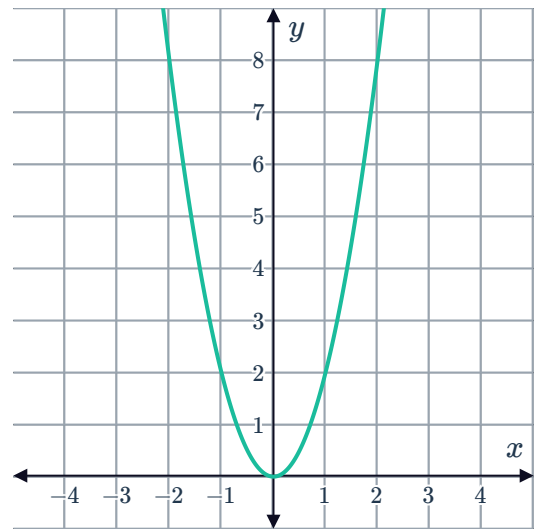
b



c



d



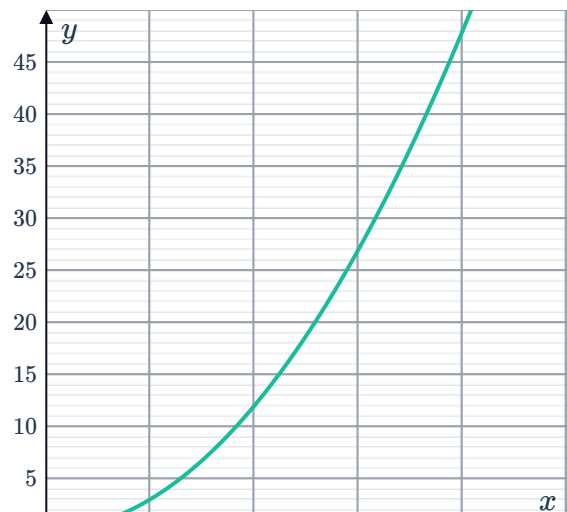
Applications

14

a

x (m)	0	1	2	3	4
y (m)	0	3	12	27	48

b



c 0 m

e (0,0)



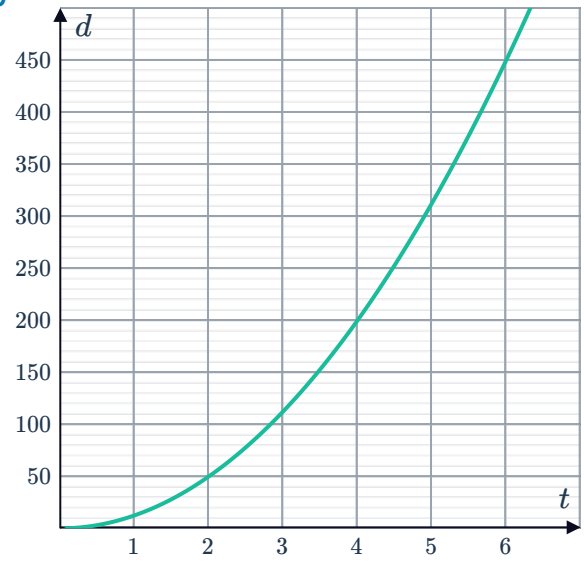
d $x = 0$

15

a

time (t)	0	2	4	6
distance (d)	0	50	200	450

b



c $t = 3$ s